

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

SE 19-206-0404 PLANNING AND DESIGN OF FIRE PROTECTION SYSTEMS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Explain the working principle of different types of fire detectors and alarm system and plan and distribute them in buildings of different occupancy.
- CO2: Interpret the concept of fire extinguishment with different extinguishants and the working principle and operation of portable fire extinguishers.
- CO3: Plan and distribute appropriate portable extinguishers in buildings of different occupancy.
- CO4: Design fixed sprinkler system in buildings of different occupancy as per Indian Standard Specifications.
- CO5: Interpret appropriate standards for the design and installation of fixed CO₂ and DCP systems as per Indian Standard Specifications.
- CO6: Interpret appropriate standards for the design and installation of fixed foam systems and apply these concepts in the design of typical hydrocarbon storage tank protection.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Write down the various principles used in heat detections.	3	L2	1	3
	(b)	Compare photoelectric type smoke detectors with air sampling type smoke detectors.	3	L2	1	3
	(c)	Water is most commonly used extinguishing agent. Comment.	3	L1	2	3
	(d)	Explain the method used to fight polar solvent fuel fires.	3	L2	2	3
	(e)	Compare “Sprinkler” with Drencher.	3	L1	4	3
	(f)	Explain the working of Deluge Installations.	3	L2	4	3
	(g)	List the general requirements for installation of fixed CO ₂ fire extinguishing system.	3	L1	6	3
	(h)	Calculate the free venting area for an enclosure made of normal type of construction with 10 numbers of number 6 nozzles.	3	L3	6	3

PART B

(4 × 12 = 48)

II.	(a)	Explain the working fixed temperature detector using fusible alloy.	6	L2	1	3
	(b)	Outline the performance criteria for heat sensitive fire detector.	6	L2	1	3
OR						
III.	(a)	Explain the working of ultra violet flame detector.	6	L2	1	3
	(b)	Explain the working of Pellister type gas detectors.	6	L2	1	3

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Explain the concept of firefighting with powders.	6	L2	2	3
	(b) Compare fluoro protein foam concentrate with aqueous film forming concentrate.	6	L2	2	3
OR					
V.	(a) Explain the concept of firefighting with inert gases.	6	L2	2	3
	(b) Explain the any three first aid fire protection measures.	6	L2	3	3
VI.	(a) Summarise the working of high velocity and medium velocity spray system.	6	L2	4	3
	(b) Outline the working of fusible ink type sprinkler heads.	6	L2	4	3
OR					
VII.	(a) Explain the components and working of quartzoid bulb type sprinkler head.	6	L2	4	3
	(b) Summarise the design aspects of water sprinkler system as per relevant BIS.	6	L2	4	3
VIII.	With neat sketch explain total flooding and local application systems. Explain the use of total flooding and local application system in two industrial applications.	12	L2	6	3
OR					
IX.	Define a "tank farm". Which are the various firefighting requirements for a tank farm. Explain. List and explain the various firefighting facility in a class B storage tank.	12	L2	6	3

Bloom's Taxonomy Levels

L1 - 7.5%, L2 - 90%, L3 - 2.5%.
